42. DEVELOPMENT OF THE LIMBS

DURATION : 10:30 STUDY SHEET

COURSE SUMMARY

This video highlights the fascinating link between limb development and internal structures, particularly the brain and digestive tract. You will learn how childhood injuries can influence current pain and how approaches in Chinese medicine and osteopathy can be integrated to treat tendinitis and other joint pain. Key concepts discussed include peritoneal dynamics, the importance of blood vessels in limb formation, and the interaction between bones and muscles during development. By understanding these complex relationships, you will be better equipped to effectively treat limb injuries and improve the overall health of your patients.

LIMB DEVELOPMENT

Limb development occurs in parallel with internal structures, particularly the **brain** and the **digestive tract**. After rotation, the lower and upper limbs form simultaneously.

Tendonitis can reflect a deficit in the rotation of the **peritoneum**. For example, a person experiencing wrist pain, often diagnosed as **tennis elbow**, may not have played the sport. It is crucial to explore **pulmonary** or **visceral** aspects to understand the origin of the problem. By analysing the organisation of the system, one can treat the visceral to free the limbs.

In Chinese medicine, specific peripheral points are stimulated to treat internal problems. The imprint of the peritoneum can reactivate to reuse previous functions. Major life injuries often occur before the age of 5, and some can even date back to the embryonic stage. What happens in the body can be reflected in the limbs.

The **cellulome** and the **neural tube** play an essential role in development. A specific angle creates a suction field, influencing growth. A metabolic field with areas of restriction and overgrowth leads to differential growth movements, influenced by the vascular system. Active metabolism generates a powerful flow in the **mesenchymal** tissue, promoting the growth of **ectodermal** tissue.

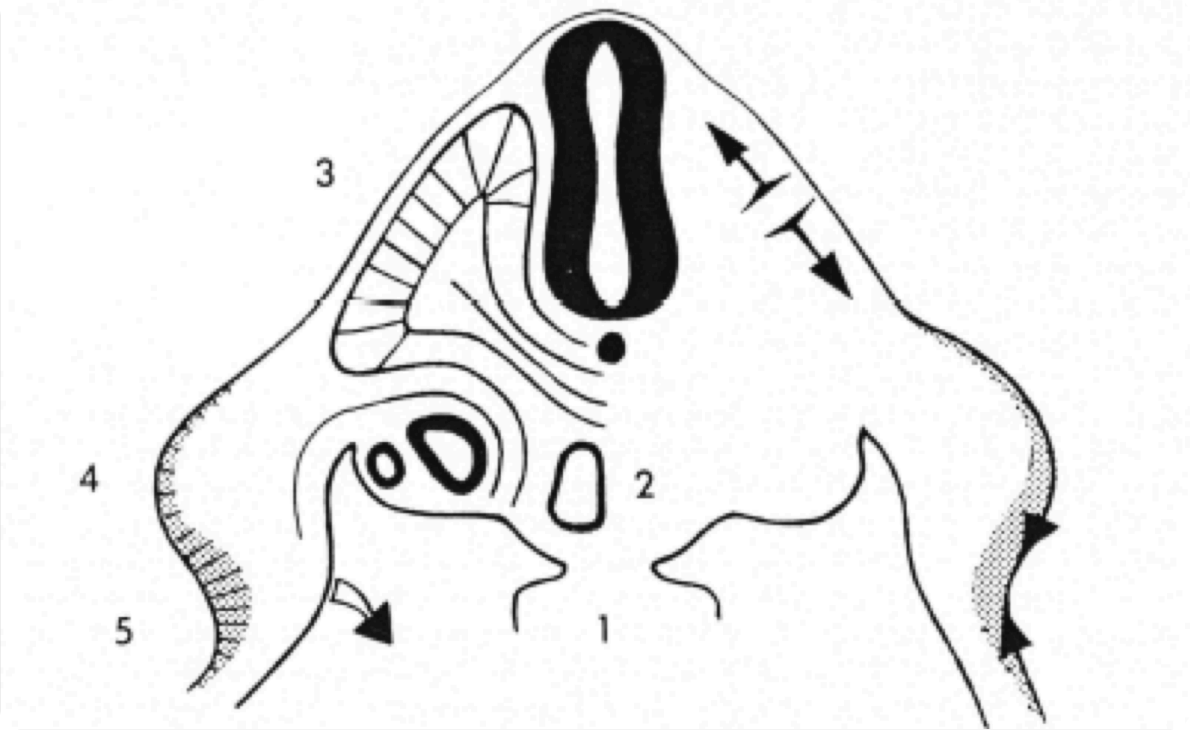


FIG. — Concept of aspiration and restriction fields, logically placed after the explanation of these mechanisms in cells and the neural tube

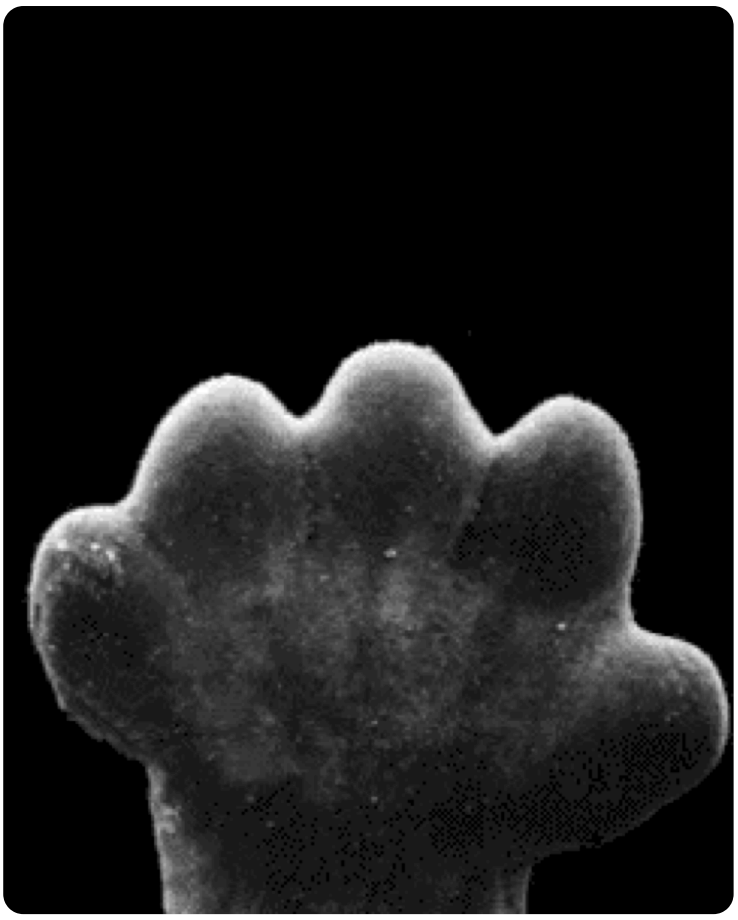


FIG. — Limb bud

The hand appears first in development, followed by the lengthening of the system. The flexion of the embryo is linked to the growth rate relative to the aorta. Joints follow a similar protocol, where the presence of blood vessels influences the density and shape of the structures. The external rotation movement is followed by an internal counter-rotation, orchestrated by the vascular field.

The **mesoblast** is responsible for the induction of the limbs, which are an expansion of the **cardio-pleuro-peritoneal sac**. Treating limb lesions requires their re-equilibration with the **pleuro-pericardio-peritoneal**. By freeing the limbs, better emergence of the upper and lower structures is allowed.

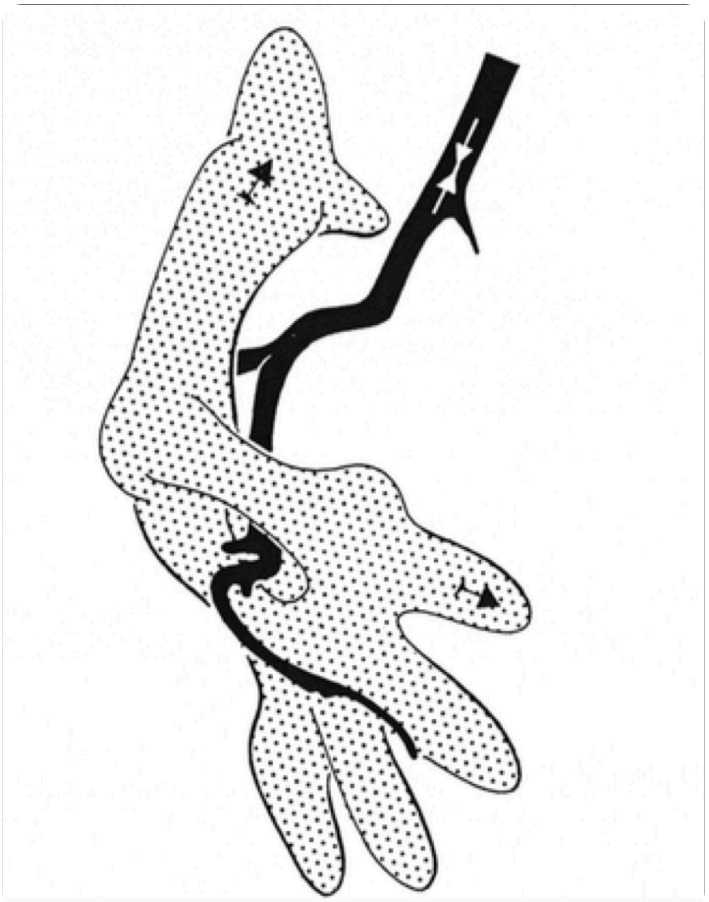


FIG. — Development of limbs

The vascular axis is fundamental for understanding metamerisation and limb arrangement. Each metamere is organised by the vessels and the neural crest, and each vertebra corresponds to an organ. Vertebral dermalgias are linked to specific organs, such as the liver or gallbladder.

The induction of the cellulome is oriented by the vascular field, influencing limb growth. The dynamic between bone and muscle is essential: initially, the bone is active, while the muscle is passive. Over time, the muscle takes over, defining the shape of the limbs.

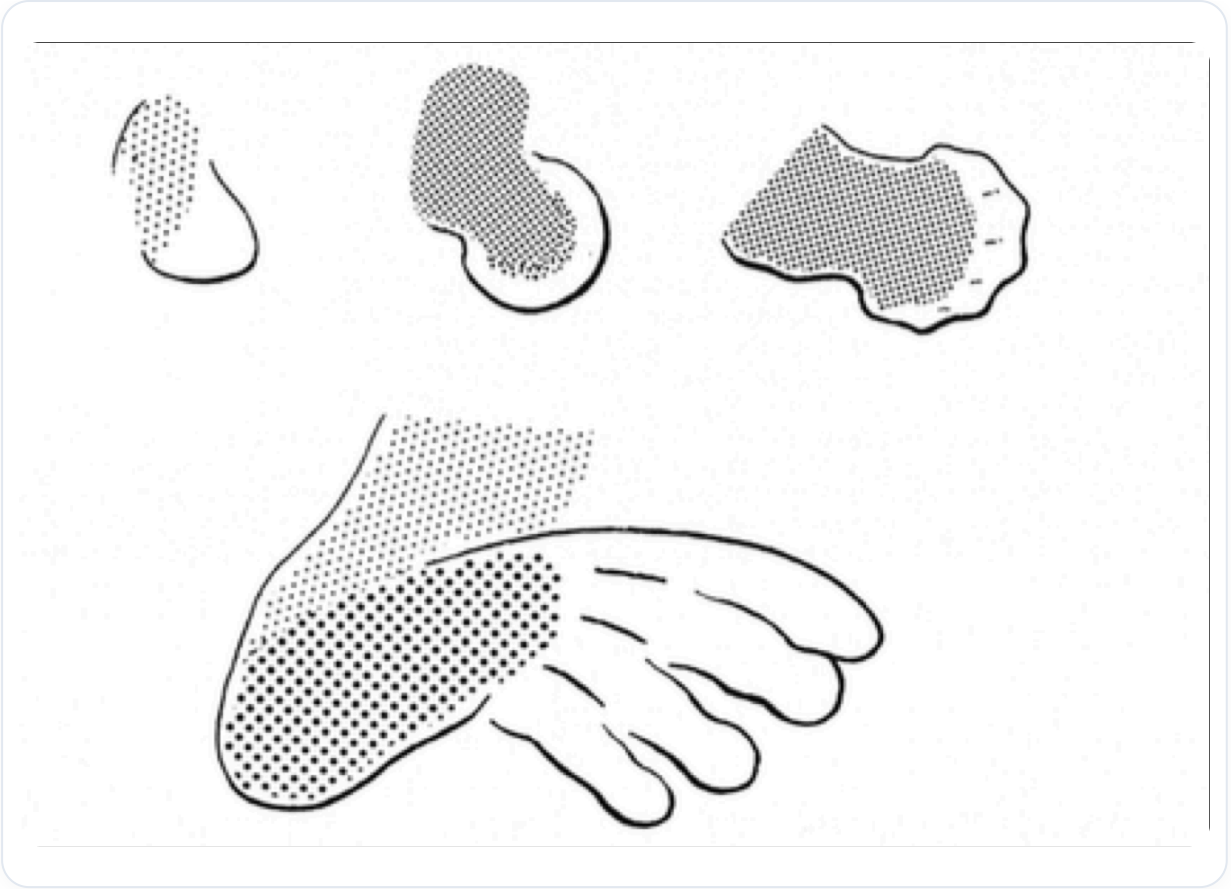


FIG. — Embryonic limb development

The development of the limbs, both upper and lower, is linked to **counter-rotation** and the organisation of the peritoneum. The base of the skull and the vertebral column derive from the **paraxial mesenchyme**, while the limb skeleton comes from the **somatopleure**.

Treating limb lesions involves reintegrating them into their peritoneal cavity, which is key to understanding the embryonic fulcrum.

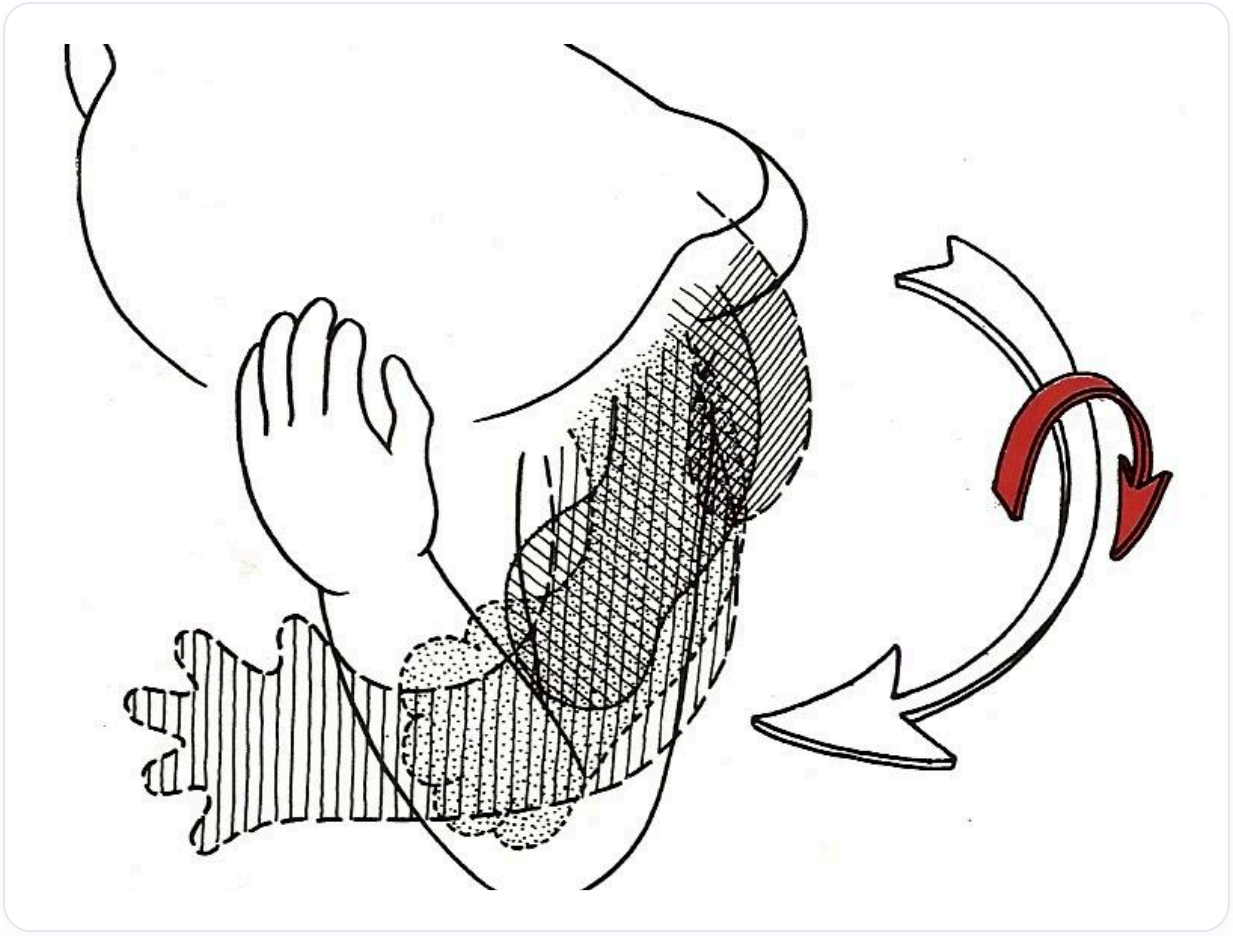


FIG. — Upper limb rotation

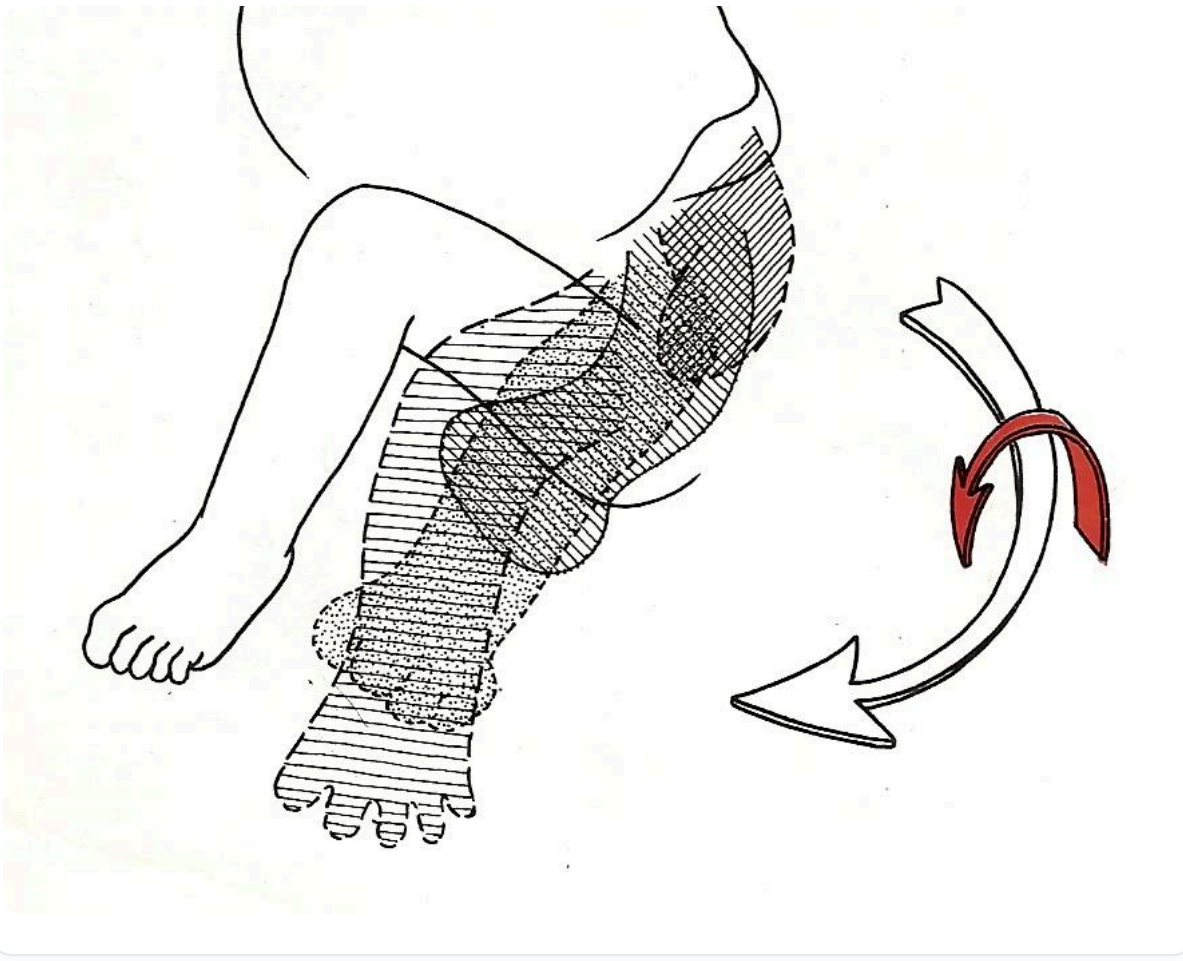


FIG. — Lower limb rotation

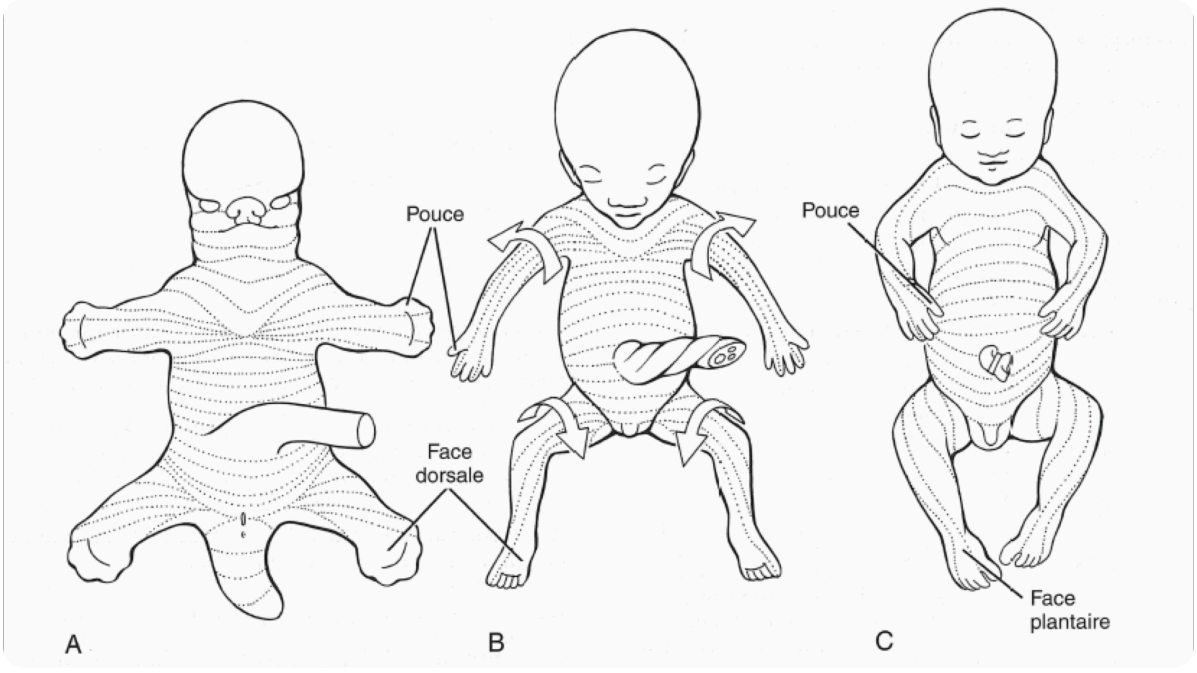


FIG. — Embryonic Langer's lines